A Semantic Transformer for Self-Supervised Deep Clustering of Large Hyperspectral Images

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***Abstract*—Local window-based vision transformers (ViTs) have recently attracted significant attention in hyperspectral image (HSI) analysis for their balance between global modeling and efficiency. However, window-based self-attention in unsupervised HSI clustering still suffers from two key limitations: 1) implicitly restricting semantic correlations to a local window, overlooking fragmented and noncontiguous yet semantically homogeneous regions; and 2) using fixed grid windows, causing boundary mixing and weakening spectral–spatial discriminative representations. To overcome these issues, we propose a semantic transformer for self-supervised deep clustering (STSDC) of large HSIs, which couples a multiscale convolutional block (MSCB) with a semantic transformer under a unified self-supervised optimization module. Specifically, the MSCB branch employs parallel receptive fields with adaptive channel recalibration to preserve fine-grained structures and suppress spectral redundancy. The semantic transformer introduces superpixel-guided attention mechanisms, including semantic contiguous attention (SCA), which selects tokens from intrasuperpixel and adjacent superpixel to enhance spatial continuity, and semantic global attention (SGA), which aggregates long-range superpixel representations to associate spatially disjoint but semantically homogeneous regions. The self-supervised module aligns the embedding representations from the MSCB and the semantic transformer via a clustering objective. Extensive experiments on benchmark HSIs demonstrate that STSDC achieves superior clustering performance compared with state-of-the-art self-supervised methods for large HSIs.**

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I. Introduction

HYPERSPECTRAL images (HSIs) capture reflectance across hundreds of contiguous bands, providing rich spectral–spatial information for fine-grained Earth observation. The rich spectral–spatial information in HSIs enables a wide range of applications, including environmental monitoring, precision agriculture, urban planning, and military reconnaissance. Among these applications, unsupervised HSI clustering is essential for discovering latent structures without requiring expensive labeled data. Specifically, clustering partitions pixels into nonoverlapping clusters based on spectral signatures and spatial context. However, HSIs are characterized by high spectral dimensionality, subtle interclass spectral differences, and irregular spatial distributions. These characteristics pose significant challenges to learning discriminative representations for clustering.

Recently, self-supervised learning (SSL) has emerged as an effective method for extracting discriminative representations from unlabeled data, significantly advancing deep clustering by improving feature separability and cluster assignment stability. In the field of HSIs, self-supervised deep clustering leverages deep neural networks to learn latent spectral–spatial embeddings from high-dimensional HSIs, thereby enabling label-free partitioning of pixels into semantically homogeneous clusters. For instance, DeepCluster alternates between feature extraction and K-means assignment, using the resulting cluster assignments as pseudo-labels to train the network, empirically enhancing feature discrimination [1]. Xie et al. [2] proposed deep embedded clustering (DEC), which employs Kullback–Leibler (KL) divergence to optimize a target distribution over clusters in the learned embedding space, thereby coupling representation learning with clustering objective refinement in a unified optimization. Building on DEC, IDEC incorporates an autoencoder (AE) reconstruction loss to preserve local structures and prevent feature degradation caused by the clustering objective [3]. More recently, contrastive learning has enhanced clustering by enforcing instance discrimination and promoting cluster compactness. For example, contrastive clustering constructs positive and negative pairs via data augmentations and jointly performs instance-level and cluster-level contrastive learning, enabling end-to-end representation learning and cluster assignment within a unified objective [4]. SwAV performs prototype-based online

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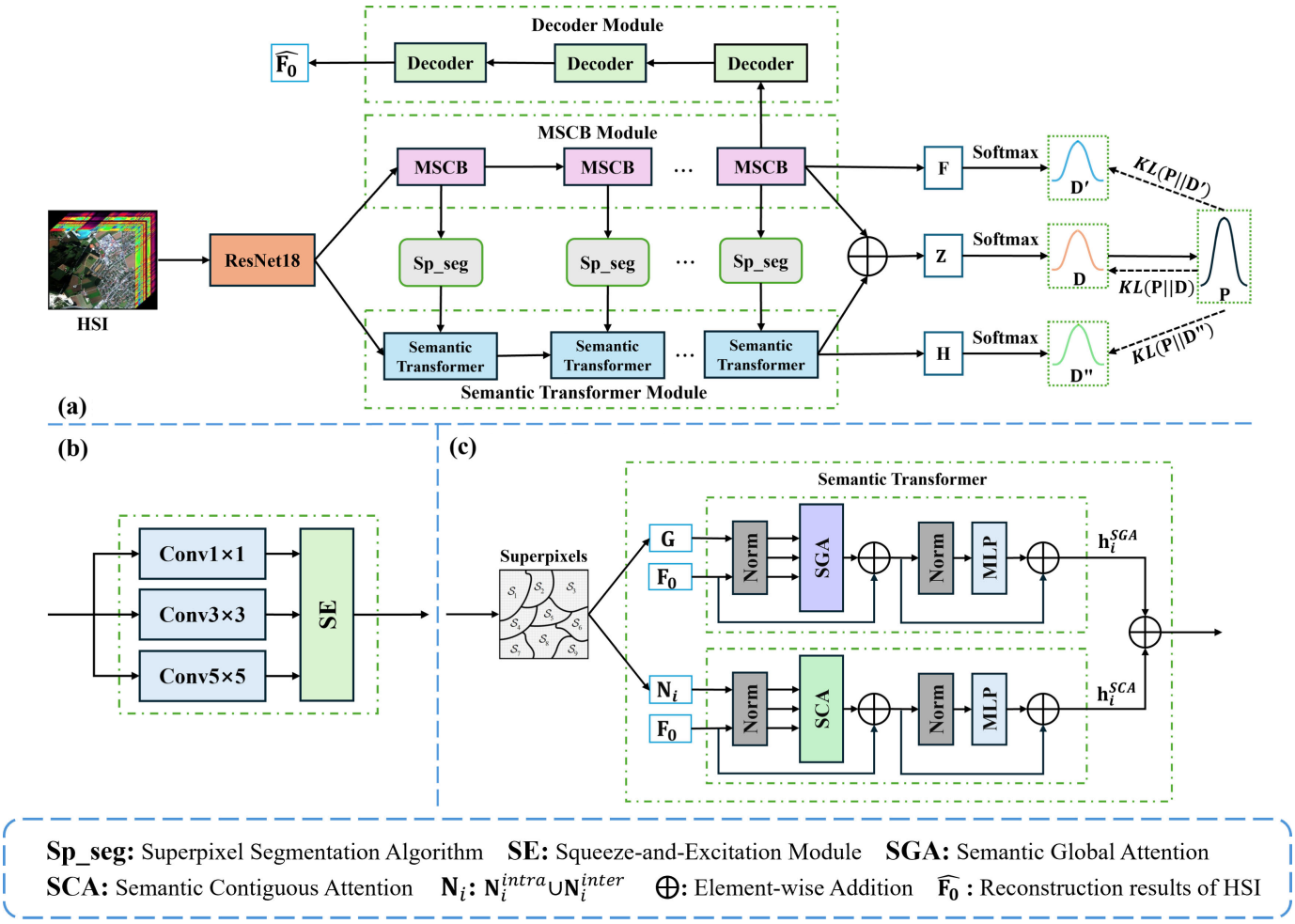


Fig. 1. Overview framework of the proposed STSDC. The STSDC consists of a ResNet-18 backbone for initial feature extraction, followed by two parallel branches: MSCB with adaptive channel recalibration for fine-grained local structure, and the semantic transformer module employs superpixel-guided SCA and SGA to aggregate features from both contiguous and disjoint semantic regions. The outputs from SGA and SCA are fused and optimized by a self-supervised module aligning local and global embeddings via clustering objectives. The Decoder module reconstructs the input features from the MSCB branch and introduces a reconstruction loss, which stabilizes training, preserves local structural details, and prevents feature degradation during clustering optimization. The STSDC efficiently bridges fine-grained spatial structures and global semantic consistency for unsupervised clustering in large HSIs. (a) STSDC. (b) MSCB. (c) Semantic transformer.

clustering across multiple augmented views and enforces cluster assignment consistency through a swapped-prediction mechanism, enabling scalable representation learning without explicit pairwise contrast [5].

Most existing self-supervised deep clustering frameworks for HSIs adopt convolutional neural networks (CNNs) as backbones. Although CNNs extract local spatial features effectively, fixed kernels and predefined receptive fields restrict the ability to model long-range spectral–spatial dependencies. In contrast, vision transformers (ViTs), driven by self-attention mechanisms, have demonstrated significant potential in advancing HSI analysis. Compared with CNNs, ViTs can more effectively capture complex spectral and spatial relationships by leveraging long-range dependencies. Representative transformer-based architectures include the original transformer encoder [6], DETR-style designs [7], and hybrid frameworks such as TNT [8]. However, traditional fully connected self-attention suffers from quadratic computational complexity with respect to spatial resolution, making it impractical for processing large HSIs with high spatial resolution.

To alleviate the quadratic computational complexity posed by fully connected self-attention, window-based ViTs partition the image into fixed local windows and perform self-attention within each window, significantly reducing computational complexity. Representative methods include Swin transformer [9], PVT [10], shuffle transformer [11], and HaloNet [12]. To facilitate cross-window information interaction, CSWin transformer computes self-attention along cross-shaped windows in horizontal and vertical directions [13]. RegionViT leverages region partitioning and intraregion attention to optimize computational efficiency [14]. Twins transformer combines local-window attention with a global subsampling mechanism to strengthen global modeling [15]. In HSIs, semantically homogeneous regions can be fragmented and noncontiguous. However, local window-based ViTs constrain each query to interact only with tokens inside a predefined local window, which makes it difficult to capture dependencies among

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noncontiguous or irregularly shaped semantic regions. Consequently, restricting self-attention to fixed windows introduces two fundamental limitations for HSIs: 1) it tends to overfit local structures that may be semantically heterogeneous, leading to the mixing of features across region boundaries; and 2) it inadequately models long-range semantic dependencies across spatially disjoint yet spectrally similar regions, which diminishes intercluster separability. In practice, the above fundamental limitations show up as specific challenges in self-supervised HSI clustering. First, relying on spatial adjacency rather than semantic similarity degrades boundary integrity. Specifically, aggregating adjacent but semantically dissimilar pixels causes feature mixing at boundaries, resulting in blurred cluster assignments. Second, the inability to capture long-range interactions across fragmented but spectrally similar regions affects the formation of clear clusters across spatially disjoint regions and ultimately degrades clustering performance.

To address the aforementioned challenges, we propose a semantic transformer for self-supervised deep clustering (STSDC) of large HSIs, which exploits semantic structure efficiently by allowing each query to attend to key–value pairs associated with semantic regions. The semantic transformer, guided by semantic homogeneous regions, enables queries to aggregate information from both spatially adjacent and disjoint regions sharing similar spectral properties, enhancing intercluster separability without relying on predefined window partitions. In addition, by explicitly modeling superpixels as semantic primitives, STSDC preserves boundary integrity and captures long-range dependencies while maintaining near-linear complexity relative to HSI resolution. Traditional ViTs face computational challenges on high-resolution HSIs due to fully connected self-attention with quadratic complexity. Meanwhile, window-based variants reduce complexity via fixed grids, predefined directions, or hierarchical partitions. Instead, STSDC constructs superpixel-level key–value pairs to compute semantic attention rather than relying on rigid spatial proximity or fixed geometric partitioning. As depicted in Fig. 1, STSDC comprises three principal components: a multiscale convolutional block (MSCB), a semantic transformer, and a self-supervised optimization module. Specifically, a ResNet backbone first extracts general low-level representations, which are then processed in parallel by a dual-branch architecture that leverages the complementary strengths of the MSCB and the semantic transformer. The MSCB branch emphasizes fine-grained local details, whereas the semantic transformer branch aggregates semantically homogeneous context across spatial adjacent and fragmented regions. To overcome the limitations of window-based ViTs, semantic transformer introduces two complementary attention mechanisms. Semantic contiguous attention (SCA) employs superpixel-guided token selection to aggregate features from intrasuperpixel and adjacent regions, enhancing spatial continuity. Semantic global attention (SGA) captures long-range dependencies across spatially disjoint yet spectrally similar regions via superpixel-level representations. By jointly optimizing the dual branches within a self-supervised module, STSDC learns semantically homogeneous embeddings and enhances cluster separability by aligning latent representations between the MSCB and the semantic transformer.

The main contributions are summarized as follows.

1) We propose STSDC, a unified self-supervised deep clustering framework for large HSIs that jointly trains an MSCB and a semantic transformer via a self-supervised optimization module, where the MSCB with adaptive channel recalibration preserves fine-grained structures and multiscale context information, and the superpixel-guided semantic attention efficiently aggregates features from semantically homogeneous regions, including both spatially adjacent and disjoint regions, with lower computational cost. STSDC simultaneously preserves local structures and maintains global semantic consistency, producing semantically homogeneous embeddings.
2) To overcome the boundary feature mixing and limited cross-region interactions caused by fixed window-based self-attention, we propose a superpixel-guided semantic transformer that leverages superpixels as region-level priors to couple local continuity with long-range semantics. Specifically, SCA selects superpixel-guided intraregion and adjacent-region tokens to enhance spatial continuity. Meanwhile, SGA aggregates long-range superpixel representations to associate spatially disjoint but semantically homogeneous regions. Together, these mechanisms jointly improve intracluster compactness and intercluster separability.
3) Extensive experiments on large benchmark HSIs demonstrate that the proposed STSDC framework outperforms state-of-the-art self-supervised deep clustering methods in terms of clustering accuracy, validating the effectiveness of STSDC.

II. Related Work

A. Deep Clustering

Deep clustering combines the powerful feature extraction capabilities of deep neural networks with traditional clustering algorithms to discover semantic structures in unlabeled high-dimensional data. Existing methods can be broadly grouped into AE-based deep clustering, variational AE (VAE)-based deep clustering, generative adversarial network (GAN)-based deep clustering, and contrastive learning-based deep clustering.

1) *AE-Based Deep Clustering:* AE-based deep clustering methods learn compact latent representations via an encoder–decoder architecture optimized by a reconstruction loss, while introducing an auxiliary clustering objective to promote cluster-friendly structure in the latent space. Representative methods include DEC and IDEC. Furthermore, Dizaji et al. [16] proposed a deep clustering framework that jointly embeds convolutional AEs with relative entropy minimization. In this framework, the reconstruction loss of the AE is used as a data reconstruction regularization term, which effectively prevents the deep embedding function from overfitting. However, AE-based deep clustering methods are

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commonly highly sensitive to initialization, which may lead to suboptimal partitions.

2) *VAE-Based Deep Clustering:* VAE-based deep clustering methods introduce explicit probabilistic modeling assuming latent variables are drawn from parameterized distributions, which are then optimized via variational inference. VaDE [17] integrates a Gaussian mixture prior into a VAE, enabling the joint optimization of cluster assignments and the generative model through variational inference. GMVAE [18] generalizes mixture modeling in the latent space via a VAE and mitigates cluster degeneracy with a minimum information constraint. Yang et al. [19] proposed DGG, which integrates a Gaussian mixture VAE with graph embedding, and leverages local structural information and global mixture modeling to jointly optimize representation learning and cluster assignments.
3) *GAN-Based Deep Clustering:* GAN-based deep clustering methods leverage adversarial training between a generator and a discriminator to learn data distributions in which latent variables encode cluster structure. InfoGAN [20] learns disentangled and interpretable latent representations by maximizing mutual information via a variational lower bound. ClusterGAN [21] combines discrete and continuous latent variables with a clustering-specific loss to jointly train a GAN and an inference network, achieving cluster-separable latent representations. ALRDC [22] injects adversarial perturbations into unsupervised representation learning to enhance robustness and cluster separability.
4) *Contrastive Learning-Based Deep Clustering:* Contrastive learning has recently shown effectiveness in unsupervised representation learning and deep clustering. By maximizing similarity between augmented views of the same instance (positive pairs) and minimizing similarity with different instances (negative pairs), contrastive learning produces discriminative representations that help partition the data into clusters. Xu et al. [23] proposed a graph-based consistency learning for contrastive multiview clustering, which improves cross-view alignment and clustering performance through cross-view contrast and graph structural consistency constraints. Xu et al. [24] proposed the GLAC-GCN framework, which combines global semantic information and local topological structure within a graph convolutional network (GCN), and optimizes the combined objectives with contrastive losses to enhance structure-aware graph clustering. Liu et al. [25] proposed simple contrastive graph clustering, which uses lightweight contrastive objectives to optimize graph representations and reduces computational cost while maintaining competitive clustering accuracy. Zhang et al. [26] proposed MSCSC, which combines masked superpixel contrastive learning and subspace clustering to enhance feature learning and clustering accuracy for HSIs. Guan et al. [27] proposed contrastive multiview subspace clustering for HSIs using GCNs, which leverages spatial–spectral and textural information to learn affinity matrices and employs contrastive learning to enhance feature robustness and consistency across views. Luo et al. [28] proposed EMVCC, which utilizes spatial and spectral features with a joint self-supervised loss to minimize false negative pairs and align representations with high-confidence pseudo-labels. Yao et al. [29] proposed HCL-Net for large-scale HSIs, which utilizes a diffusion hypergraph to model high-order correlations and a confidence-guided strategy to dynamically update positive and negative pairs, enhancing clustering performance. Yang et al. [30] proposed the S2GCL framework for robust HSI clustering, which designs a multilayer perceptron (MLP) to generate low-dimensional embeddings instead of GCNs. Qi et al. [31] proposed the SPGCC, which extracts high-order features and uses semantic-preserving data augmentations to enhance clustering performance.

B. Vision Transformer

ViTs have achieved remarkable success in computer vision. By treating image patches as sequence tokens, ViTs leverage multihead self-attention (MHSA) to capture long-range dependencies. For an input sequence of N patch tokens, MHSA forms an $N \times N$ attention matrix per head. The computation complexity is $O(N^2C)$ for computing $\mathbf{QK}^T$ and multiplying by $\mathbf{V}$, and the memory complexity is $O(N^2)$ for storing attention matrices. The quadratic cost of fully connected self-attention constrains training and inference on large HSIs.

To address the above limitations, researchers have proposed sparse attention, linear attention, and attention compression to reduce computation and memory consumption. Sparse attention constrains self-attention to local partitions (e.g., windows or stripes), and uses designed cross-window mechanisms to exchange information across different regions. Specifically, Swin transformer [9] partitions tokens into nonoverlapping windows, applies window-based self-attention, and uses shifted windows within a hierarchical pyramid to exchange cross-window information. CSWin [13] forms cross-shaped stripes along horizontal and vertical axes and computes self-attention within stripes, capturing long-range context. Liu et al. [32] proposed a spectral Swin transformer network for HSIs, which employs group attention and sliding window attention mechanisms to enhance feature representation and retain global contextual information. RegionViT [14] constructs hierarchical regional tokens by aggregating local patches and achieves multiscale modeling via explicit interactions between regional and local tokens. MSG-transformer [33] allocates one messenger token for each region to mediate interregion communication. DAT [34] leverages deformable self-attention to focus on semantically relevant spatial regions, dynamically aligning features and enhancing representation learning. Although sparse attention mechanisms reduce the computational and memory cost of modeling long-range dependencies in HSIs, they often require deeper stacks to propagate global semantics. In addition, the local window-based sparse attention is highly sensitive to window or stripe hyperparameters.

Linear ViTs reduce the quadratic cost of standard softmax attention by approximating the softmax kernel with

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feature maps. Kernelized linear attention rewrites the attention mechanism via associative reordering from $\text{softmax}(QK^T)V$ to $(\phi(Q)(\phi(K)^T V))$, where $\phi(\cdot)$ approximates or replaces the softmax kernel, reducing the complexity to $O(N)$. Performer [35] introduces FAVOR+, using positive orthogonal random features to provide an unbiased softmax-kernel approximation while better controlling the estimation variance, achieving linear complexity while remaining fully compatible with standard transformers. EcoFormer [36] uses kernelized binary embeddings for queries and keys to approximate attention with linear complexity, achieving energy-efficient computation and reduced memory consumption. Kernel variants such as Kerformer [37] use deterministic kernel feature maps for numerically stable softmax-like approximations.

Attention compression methods reduce the size of attention (typically key/value) or reorganize tokens to simplify the attention matrix and improve computational efficiency. MaxViT [38] employs multiaxis attention, combining blocked local and dilated global attention, to efficiently capture both short- and long-range dependencies. PVTv2 [39] uses spatial-reduction attention that downsamples key/value to reduce the number of tokens. ConvFormer [40] integrates convolutional operators into attention blocks to reduce effective token resolution and strengthen local inductive bias. Linformer [41] assumes that the attention matrix is approximately low rank and learns projections that compress key/value into a low-dimensional subspace, reducing the number of tokens. Luo et al. [42] proposed SDST for HSI clustering, which combines a shared autoformer for global feature learning with a Siamese transformer module to enhance superpixel-level representation. Xu et al. [43] proposed DSFormer, which jointly models spatial and spectral contexts by flexibly selecting and fusing features across different receptive fields, effectively reducing irrelevant information interference by focusing on the most relevant spatial–spectral tokens of HSIs.

III. Proposed Method

A. Overall Framework

As illustrated in Fig. 1, we propose an STSDC of large HSIs, which integrates an MSCB and a semantic transformer, optimized jointly under a self-supervised module. Specifically, given an input HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, where H denotes the height, W denotes the width, and C denotes the spectral channels, a ResNet backbone first extracts shallow spatial–spectral features $\mathbf{F}_0 \in \mathbb{R}^{H\times W\times D_0}$, where D_0 denotes the feature dimension. The shallow features are then processed in parallel by a dual-branch architecture: 1) the MSCB branch, designed to capture fine-grained local structures across multiscale receptive fields with adaptive channel recalibration to suppress spectral redundancy; and 2) the semantic transformer branch, which leverages superpixel-guided attention mechanisms to overcome the limitations of fixed window-based self-attention. To address boundary mixing and the lack of cross-region semantic interactions in window-based ViTs, the semantic transformer introduces two complementary attention mechanisms: SCA selects superpixel-guided intraregion and adjacent-region tokens to enhance spatial continuity, while SGA aggregates long-range superpixel representations to associate spatially disjoint but semantically homogeneous regions. Finally, the end-to-end STSDC framework is optimized through a self-supervised module, aligning the embedding representations from both branches via a clustering objective to enhance cluster separability.

B. MSCB With Adaptive Channel Recalibration

HSIs are characterized by diverse object scales, irregular geometric structures, and complex spectral–spatial structures. Traditional single-scale convolutions are often inadequate for capturing both fine-grained local details and global contextual information. Specifically, large-scale convolutional kernels tend to over-smooth small-scale structures, while small-scale convolutional kernels may fail to incorporate sufficient global context. To overcome these limitations, we propose an MSCB, which employs parallel receptive fields with adaptive channel recalibration to preserve fine-grained structures and suppress spectral redundancy.

Specifically, given the input feature $\mathbf{F}_0 \in \mathbb{R}^{H\times W\times D_0}$, with a set of kernel sizes defined as $\mathcal{K} = \{k_1, \ldots, k_n\}$. For the ith scale, where $i \in \{1, \ldots, n\}$, a convolution operation with kernel size k_i is performed using learnable weights $\mathbf{W}_{k_i}$ and bias $\mathbf{b}_{k_i}$. A nonlinear activation function $\psi(\cdot)$ is then applied to enhance the representation capacity. Finally, the outputs from all scales are concatenated along the channel dimension to construct a unified multiscale representation

$$\mathbf{F}^{cat} = \text{Concat}_{k_i\in\mathcal{K}}\ \psi\left(\mathbf{W}_{k_i} * \mathbf{F}_0 + \mathbf{b}_{k_i}\right) \tag{1}$$

where * denotes the convolution operation.

In HSIs, feature channels encode fine-grained spectral signatures. To enhance spectral interactions and reduce spectral redundancy, the squeeze-and-excitation (SE) block is employed for adaptive channel recalibration. Specifically, the recalibration begins with global average pooling, defined as $\zeta_c = (1/H \times W)\sum_{i=1}^{H}\sum_{j=1}^{W} f_c^{cat}(i, j)$, which aggregates spatial–spectral features into a compact descriptor ζ_c for the cth channel, where $f_c^{cat}(i, j)$ denotes the feature value at spatial position (i, j) of the cth channel in the $\mathbf{F}^{cat}$. These descriptors form a channel vector $\boldsymbol{\zeta} = [\zeta_1, \ldots, \zeta_c, \ldots, \zeta_C]$, where C is the total number of channels. Channel interdependencies are then modeled using a nonlinear gating mechanism with two fully connected layers $\mathbf{s} = \sigma(\mathbf{W}_2\delta(\mathbf{W}_1\boldsymbol{\zeta}))$, where $\mathbf{W}_1$ and $\mathbf{W}_2$ are weight matrices for dimensionality reduction and expansion, $\delta(\cdot)$ is the ReLU activation function, and $\sigma(\cdot)$ is the sigmoid function. The resulting channel-wise weights $\mathbf{s} = [s_1, \ldots, s_C]$ assign importance to each spectral channel. Let $\mathbf{f}_i^{cat} \in \mathbb{R}^C$ denote the spectral feature corresponding to the ith feature of $\mathbf{F}^{cat}$. We obtain the refined feature $\mathbf{f}_i$ by applying an element-wise multiplication between the channel weights $\mathbf{s}$ and $\mathbf{f}_i^{cat}$, such that $\mathbf{f}_i = \mathbf{s} \odot \mathbf{f}_i^{cat}$, where $\odot$ denotes the element-wise product. Consequently, $\mathbf{F} = [\mathbf{f}_1, \ldots, \mathbf{f}_i \ldots, \mathbf{f}_N]$ denotes feature representation from the MSCB branch, where $N = H \times W$ is the total number of pixels. By amplifying informative channels and suppressing redundant ones, the recalibration mechanism improves the discriminative capacity of spectral representations in high-dimensional HSIs.

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C. Semantic Transformer for Large HSIs

In HSIs, object regions are characterized by irregular geometric structures, fragmented spatial distributions of similar objects, and intricate region boundaries. Traditional fixed window-based self-attention mechanisms tend to aggregate heterogeneous features across boundaries, resulting in feature mixing across region boundaries. Moreover, they fail to associate spatially disjoint but spectrally similar regions, reducing intercluster separability. To address these limitations, we propose a semantic transformer that leverages superpixel-level spatial priors to guide semantic feature learning. The semantic transformer consists of SCA, SGA, and a feature enhancement module.

1) Superpixel Segmentation-Based SCA: Existing window-based ViTs compute self-attention by restricting interactions between each query and its adjacent keys within a fixed local window, capturing contextual dependencies. However, such windows are typically defined by grid partitioning, which fails to align with object boundaries, thereby disrupting spatial continuity and semantic consistency.

To address these shortcomings, we propose the SCA mechanism, which utilizes a superpixel-guided token selection strategy to aggregate features within spatially continuous regions and their adjacent neighborhoods. Specifically, SCA replaces fixed grid windows with adaptive regions derived from entropy rate superpixel (ERS) segmentation, selecting keys and values from spectrally similar and spatially continuous pixels within a superpixel, complemented by semantically homogeneous adjacent regions. By restricting attention to intrasuperpixel pixels and semantically homogeneous adjacent regions, SCA suppresses cross-boundary mixing, improves spatial continuity, and minimizes redundant correlations across heterogeneous regions. To introduce explicit spatial semantic priors, the ERS segmentation algorithm is first applied to the feature map $\mathbf{F}$, generating M nonoverlapping superpixels $\{\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_M\}$. Each feature $\mathbf{f}_i$ from $\mathbf{F}$ is uniquely assigned to a specific superpixel $\mathbf{S}_m$, where $m \in \{1, \ldots, M\}$. ERS prefers compact regions with strong internal cohesion and clearly delineated boundaries, which we exploit as label-free semantic proxies. Crucially, the ERS objective maximizes the entropy rate under boundary regularization, ensuring that each superpixel maintains high internal spectral similarity and aligns with object boundaries. In HSIs, pixels are commonly spectrally smooth within local neighborhoods and exhibit abrupt transitions across boundaries. ERS is well-matched to these characteristics by enforcing strong intrasuperpixel similarity to capture homogeneous spectral and textural information, placing cuts at spectral or gradient discontinuities to align with semantic boundaries, and promoting spatial compactness to preserve contiguous object support and reduce fragment mixing. Consequently, ERS-based superpixels serve as effective region-level semantic representations, approximating homogeneous pixels in HSIs while preserving spatial continuity.

During the transformer encoding stage, a superpixel-guided key/value construction strategy is introduced to ensure that self-attention focuses on semantically homogeneous regions. The strategy balances the fine-grained details within superpixel regions and the contextual information across adjacent regions. Specifically, for $\mathbf{f}_i \in \mathbf{S}_m$. Based on $\mathbf{S}_m$, an intrasuperpixel feature vector $\mathbf{N}_i^{intra}$ is constructed by selecting the j nearest features of $\mathbf{f}_i$ within the superpixel $\mathbf{S}_m$ by K-nearest neighbors algorithm. The intrasuperpixel feature is expressed as $\mathbf{N}_i^{intra} = [\mathbf{f}_{i,1}, \ldots, \mathbf{f}_{i,j}]$, where $\mathbf{f}_{i,j} \in \mathbb{R}^C$.

In addition to modeling intrasuperpixel characteristics, contextual information from adjacent superpixels is incorporated to capture intersuperpixel dependencies. Adjacent superpixels are defined based on spatial proximity, where two superpixels $\mathbf{S}_m$ and $\mathbf{S}_j$ are considered adjacent if they share a common boundary. For each adjacent superpixel $\mathbf{S}_j$, a superpixel representation $\mathbf{r}_j$ is computed as the mean feature vector of all pixels within $\mathbf{S}_j$. To prevent introducing heterogeneous context and focus on the most relevant cross-region information, we retain only the top-k adjacent superpixel representations that are most similar to the $\mathbf{f}_i$, where $\mathbf{f}_i$ denotes a feature within $\mathbf{S}_m$, forming $\mathbf{N}_i^{inter} = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_k]$. Specifically, we measure the feature similarity (e.g., cosine similarity) between the pixel $\mathbf{f}_i$ and the superpixel representation $\mathbf{r}_k$ from the adjacent superpixel of $\mathbf{S}_j$, retaining only the top-k most similar superpixel representations. This ensures uniform tensor dimensions during attention computation while emphasizing the most significant contextual information from adjacent superpixels.

By combining the intrasuperpixel representations and cross-region superpixel representations, we construct the key and value sets for the query $\mathbf{q}_i$ as follows:

$$\mathbf{K}_i = \mathbf{V}_i = \mathbf{N}_i^{intra} \cup \mathbf{N}_i^{inter} = \begin{bmatrix} \mathbf{f}_{i,1}, \mathbf{f}_{i,2}, \ldots, \mathbf{f}_{i,j} \\ \mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_k \end{bmatrix}. \tag{2}$$

The constructed key and value sets are then transformed into query–key–value via learnable linear projections

$$\begin{aligned} \mathbf{Q}^{SCA} &= \mathbf{F}_0 \mathbf{W}_Q^{SCA} \\ \mathbf{K}_i^{SCA} &= \mathbf{K}_i \mathbf{W}_i^{SCA} \\ \mathbf{V}_i^{SCA} &= \mathbf{V}_i \bar{\mathbf{W}}_i^{SCA} \end{aligned} \tag{3}$$

where $\mathbf{W}_Q^{SCA}$, $\mathbf{W}_i^{SCA}$, and $\bar{\mathbf{W}}_i^{SCA}$ are learnable projection matrices. For each query $\mathbf{q}_i$ from $\mathbf{Q}^{SCA}$, the semantic contiguous self-attention is restricted to the constructed sets for $\mathbf{K}_i$ and $\mathbf{V}_i$, ensuring that attention is focused on semantically relevant features. The SCA is computed as

$$\mathbf{h}_i^{SCA} = \text{softmax}\left(\frac{\mathbf{q}_i \left(\mathbf{K}_i^{SCA}\right)^T}{\sqrt{d_k}} \right) \mathbf{V}_i^{SCA} \tag{4}$$

where d_k denotes the feature dimension of the projected queries and keys. By restricting the key and value sets to the intrasuperpixel and cross-region representations, the SCA enables each query to focus on fine-grained features within its superpixel while incorporating semantic context from adjacent superpixels. This restriction enhances spatial continuity by preserving the semantic consistency of features within the superpixel and minimizing feature mixing across region boundaries.

2) Semantic Global Attention: In HSIs, semantically homogeneous regions often exhibit high spectral similarity even when they are spatially distant, which requires a global perspective to fully exploit these semantic relationships. Although SCA effectively enhances the consistency of features within

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contiguous semantic regions, it is inherently limited in capturing long-range semantic correlations between spatially disjoint and semantically homogeneous regions. To address this limitation, we propose SGA, which aggregates long-range superpixel representations to associate spatially disjoint and spectrally similar regions.

Specifically, for each superpixel $\mathbf{S}_m$, we compute the semantic representation via average pooling $\mathbf{g}_m = (1/|\mathbf{S}_m|)\sum_{\mathbf{f}_i \in \mathbf{S}_m} \mathbf{f}_i$, where $\mathbf{f}_i$ represents the feature vector of pixel i within $\mathbf{S}_m$, and $|\mathbf{S}_m|$ denotes the number of pixels in superpixel $\mathbf{S}_m$. The representation $\mathbf{g}_m$ captures the semantic characteristics of the superpixel region while reducing the computational burden of subsequent self-attention mechanisms, thereby enhancing scalability for large HSIs. All superpixel representations, denoted as $\mathbf{G} = [\mathbf{g}_1, \ldots, \mathbf{g}_M]$, serve as the global key/value set for the subsequent SGA. We transform $\mathbf{G}$ into query–key–value via learnable linear projections, enabling SGA to capture meaningful semantic correlations

$$\begin{aligned} \mathbf{Q}^{SGA} &= \mathbf{F}_0 \mathbf{W}_Q^{SGA} \\ \mathbf{K}^{SGA} &= \mathbf{G} \mathbf{W}_K^{SGA} \\ \mathbf{V}^{SGA} &= \mathbf{G} \mathbf{W}_V^{SGA} \end{aligned} \tag{5}$$

where $\mathbf{W}_Q^{SGA}$, $\mathbf{W}_K^{SGA}$, and $\mathbf{W}_V^{SGA}$ are learnable projection matrices. Therefore, the SGA for each query $\mathbf{q}_i$ from $\mathbf{Q}^{SGA}$ is computed as

$$\mathbf{h}_i^{SGA} = \text{softmax}\left(\frac{\mathbf{q}_i \left(\mathbf{K}^{SGA}\right)^T}{\sqrt{d_k}}\right)\mathbf{V}^{SGA} \tag{6}$$

where d_k denotes the feature dimension of the projected queries and keys. The superpixel-guided SGA enables each pixel to aggregate semantic information globally, capturing long-range dependencies between spatially disjoint but semantically homogeneous regions, thereby improving intercluster separability.

3) Feature Enhancement Module: For each query $\mathbf{q}_i$, we obtain the fine-grained feature representation $\mathbf{h}_i^{SCA}$ via SCA and the nonlocal feature representation $\mathbf{h}_i^{SGA}$ via SGA. The fused representation for $\mathbf{h}_i$ is calculated using an element-wise addition

$$\mathbf{h}_i = \mathbf{h}_i^{SCA} + \mathbf{h}_i^{SGA}. \tag{7}$$

The element-wise addition combines the strengths of SCA and SGA to produce a unified feature representation $\mathbf{h}_i$. $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_i, \ldots, \mathbf{h}_N]$ enables the model to simultaneously capture spatial continuity within adjacent regions and model long-range semantic relationships across spatially disjoint regions, jointly improving intracluster compactness and intercluster separability.

D. Self-Supervised Optimization for Latent Clustering Alignment

Building upon the dual-branch deep feature network, we propose a self-supervised optimization module designed to align and refine the latent clustering structure within the learned feature space. The self-supervised optimization module leverages the complementary strengths of the MSCB branch and the semantic transformer branch, guiding the latent representations toward a shared embedding space.

Specifically, to fully exploit the complementary spatial and semantic information, we adopt an element-wise addition $\mathbf{Z} = \mathbf{F} + \mathbf{H}$, which denotes fused feature representation, where $\mathbf{Z} = \{\mathbf{z}_1, \ldots, \mathbf{z}_N\}$. To model the clustering structure in the fused representation space $\mathbf{Z}$, we employ the heavy-tailed Student t-kernel, which is widely used in deep clustering. Using the fused feature representation $\mathbf{z}_i$ and cluster centroid $\mathbf{u}_j$, where $j \in \{1, \ldots, K\}$, we construct a soft assignment $\mathbf{P}$ as the pseudo-label

$$p_{ij} = \frac{\left(1 + \left\|\mathbf{z}_i - \mathbf{u}_j\right\|^2 \Big/ \alpha\right)^{-\frac{\alpha+1}{2}}}{\sum_{j'=1}^{K}\left(1 + \left\|\mathbf{z}_i - \mathbf{u}_{j'}\right\|^2 \Big/ \alpha\right)^{-\frac{\alpha+1}{2}}} \tag{8}$$

where p_{ij} denotes the probability that pixel i belongs to cluster j, $\mathbf{u}_j$ is initialized by applying K-means to $\mathbf{F}_0$, and α is the degrees of freedom parameter controlling the tail heaviness of the kernel. By setting $\alpha = 1$, the kernel specializes to the Cauchy kernel, which is particularly effective for the adaptive capture of clustering structures within high-dimensional fused feature spaces.

To further refine the alignment between the predictive distributions and the clustering structure, we introduce branch-specific predictive probability distributions. Let $\mathbf{D}'$, $\mathbf{D}''$, and $\mathbf{D}$ denote the predictive distributions generated by softmax applied to $\mathbf{F}$, $\mathbf{H}$, and $\mathbf{Z}$, respectively. The distributions $\mathbf{D}'$, $\mathbf{D}''$, and $\mathbf{D}$ are aligned with the target distribution $\mathbf{P}$ by minimizing the KL divergence for each branch

$$L_{KL}^{MSCB} = KL\left(\mathbf{P} \,\middle\|\, \mathbf{D}'\right) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{d'_{ij}} \tag{9}$$

$$L_{KL}^{ST} = KL\left(\mathbf{P} \,\middle\|\, \mathbf{D}''\right) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{d''_{ij}} \tag{10}$$

$$L_{KL}^{cat} = KL\left(\mathbf{P} \,\middle\|\, \mathbf{D}\right) = \sum_i \sum_j p_{ij} \log \frac{p_{ij}}{d_{ij}} \tag{11}$$

where d'_{ij}, d''_{ij}, and d_{ij} derived from $\mathbf{D}'$, $\mathbf{D}''$, and $\mathbf{D}$, respectively, denote the probability that pixel i is assigned to pseudo-label j, and p_{ij} represents the target probability computed from the d_{ij}. To ensure robust alignment and consistency across the MSCB and semantic transformer, we define the total optimization objective as the weighted sum of the KL divergences from each branch

$$L_{total} = \gamma_1 L_{KL}^{MSCB} + \gamma_2 L_{KL}^{ST} + \gamma_3 L_{KL}^{cat} + L_{res} \tag{12}$$

where γ_1, γ_2, and γ_3 are weighting coefficients to balance the contributions of each branch, and L_{res} denotes the reconstruction loss, where $L_{res} = (1/N)\|\mathbf{F}_0 - \hat{\mathbf{F}}_0\|_F^2$, which serves as an auxiliary regularizer to stabilize the training process. By imposing a reconstruction constraint, the loss encourages the network to preserve the intrinsic spectral–spatial information of HSIs, prevents unstable feature drift or collapse during early optimization, and reduces the risk of overfitting. The proposed SSL strategy aligns the local representations extracted by the MSCB with the nonlocal semantic features captured by the semantic transformer within a shared embedding space.

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The alignment unifies local and global feature distributions, improving clustering performance.

E. Computational Complexity Analysis

To demonstrate the efficiency of STSDC on large HSIs, we analyze the computational complexity of the proposed attention mechanisms compared to standard fully connected self-attention. Let $N = H \times W$ be the number of pixels, and M be the number of superpixels, where $M \ll N$. Standard fully connected self-attention requires computing similarities between all pixel pairs, resulting in a complexity of $O(N^2C)$, which is quadratic with respect to N, constituting the principal bottleneck on large HSIs.

In contrast, the proposed STSDC decouples the attention into SCA and SGA. For SCA, each pixel only attends to a limited number of samples within its superpixel and adjacent superpixels. Let t be the total number of tokens in the intrasuperpixel and adjacent superpixels, the complexity of SCA is $O(NtC)$, where $t \ll N$, resulting in linear complexity $O(N)$. For SGA, pixel-level queries attend to superpixel-level keys and values, which requires computing an $N \times M$ attention matrix with a complexity of $O(NMC)$. Therefore, the overall complexity of STSDC is approximately linear in terms of the image size N, ensuring scalability for large HSIs.

IV. Experiments

To comprehensively evaluate the effectiveness of the proposed STSDC for HSI clustering, we conduct extensive experiments on four benchmark datasets: MUUFL (ML), Pavia University (PU), Yangzhou (YZ), and Yancheng (YC) datasets. Specifically, first, we provide a detailed introduction to the HSIs and the experimental settings. Next, we compare STSDC with ten state-of-the-art deep clustering methods, highlighting its advantages. Finally, we perform ablation studies to systematically analyze the contribution of each key module.

A. Dataset Description

1) *ML Dataset:* The ML dataset was collected using a CASI-1500 imaging spectrometer over the University of Southern Mississippi Gulf Park campus in Long Beach, Mississippi, USA. It contains 64 effective spectral bands spanning wavelengths from approximately 368 to 1043 nm, with a spatial size of 325×220 pixels. The dataset is manually annotated into 11 land-cover classes, including trees, mostly grass, mixed ground surface, dirt and sand, road, water, building shadow, building, sidewalk, yellow curb, and cloth panels.
2) *PU Dataset:* The PU dataset was acquired using the ROSIS imaging spectrometer and covers the University of Pavia, Italy. The PU dataset comprises 610×340 pixels with a spatial resolution of 1.3 m and includes 103 spectral bands per pixel, spanning a wavelength range from approximately 430 to 860 nm. The PU dataset has been manually annotated into nine land-cover classes, such as asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows.
3) *YZ Dataset:* The YZ dataset was collected by the Jiangsu Provincial Engineering Research Center for Aerial Geo-detection and Intelligent Sensing using the Multi-parameter Airborne Remote Sensing System (MARSS) over YZ, China. The MARSS platform integrates a visible-light imaging spectrometer, a short-wave infrared imaging spectrometer, a RIEGL VUX-240 LiDAR sensor, and high-precision navigation systems. The dataset has a spatial size of 800×250 pixels and comprises 249 spectral bands, covering eight land-cover classes.
4) *YC Dataset:* The YC dataset was acquired on October 15, 2022, over YC City, China, from an altitude of 600 m using the MARSS equipped with a visible-spectrum imaging spectrometer. The YC dataset contains 249 spectral bands spanning wavelengths from approximately 400 to 1000 nm, with a spatial size of 400×500 pixels. The dataset provides annotations for five land-cover classes.

B. Experimental Setting

1) *Evaluation Metrics:* To comprehensively evaluate the clustering performance of the proposed STSDC, we adopt multiple evaluation metrics, including producer's accuracy (PA), user's accuracy (UA), accuracy (ACC), Fowlkes–Mallows index (FMI), adjusted Rand index (ARI), adjusted mutual information (AMI), and normalized mutual information (NMI). All metrics are computed following their standard definitions.
2) *Compared Methods:* To comprehensively evaluate the effectiveness of the proposed STSDC, we compare it with ten state-of-the-art deep clustering methods. The representative methods include DEC [2], improved DEC (IDEC) [3], deep clustering network (DCN) [44], deep embedded K-means clustering (DEKM) [45], DEC with data augmentation (DEC-DA) [46], deep clustering via weighted k-subspace network (DWSC) [47], bootstrap deep spectral clustering (BootSC) with optimal transport [48], K-AEs deep clustering (KDAE) [49], self-supervised double-structure transformer (SDST) [42], and ViT for contrastive clustering (VTCC) [50]. The configurations of these methods, including network architectures, loss functions, and optimization strategies, follow the settings described in their original papers, with minor parameter adjustments made to adapt to the HSIs.
3) *Implementation Details:* The STSDC includes a ResNet-18 backbone, a three-layer MSCB branch, a three-layer semantic transformer branch, a decoder composed of three stacked convolutional layers, and a self-supervised optimization module. The STSDC is optimized using the Adam optimizer with a learning rate of 0.005, and the training is conducted for 200 iterations. Specifically, for the ML, PU, YZ, and YC datasets, the output dimensions of ResNet-18 are set to 32, 48, 128, and 128, respectively. The convolution kernel sizes of MSCB are set to 1×1, 3×3, and 5×5, while those in the decoder are fixed to 3×3. It is worth noting that the convolution kernel weights across the stacked MSCB layers are not

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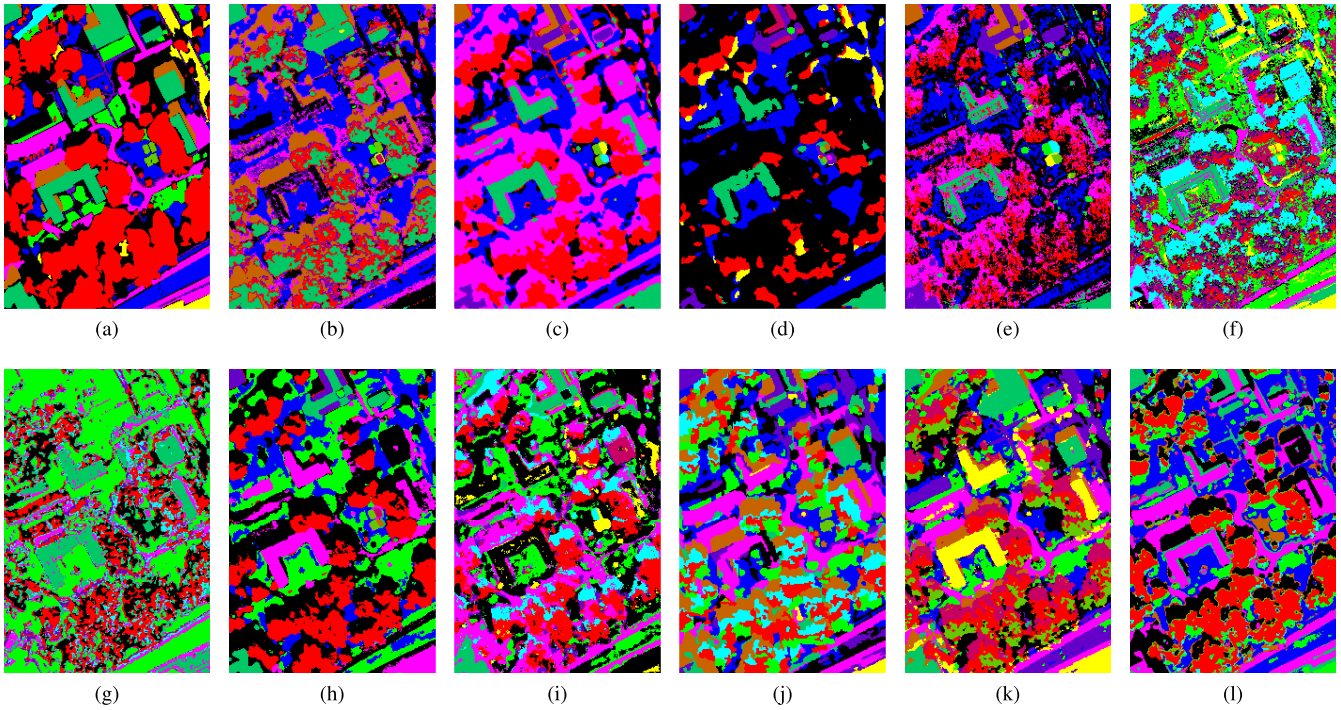


Fig. 2. Clustering maps of ML dataset. (a) Ground truth. (b) DEC. (c) DEKM. (d) IDEC. (e) KDAE. (f) DCN. (g) BootSC. (h) DEC-DA. (i) DWSC. (j) SDST. (k) VTCC. (l) STSDC.

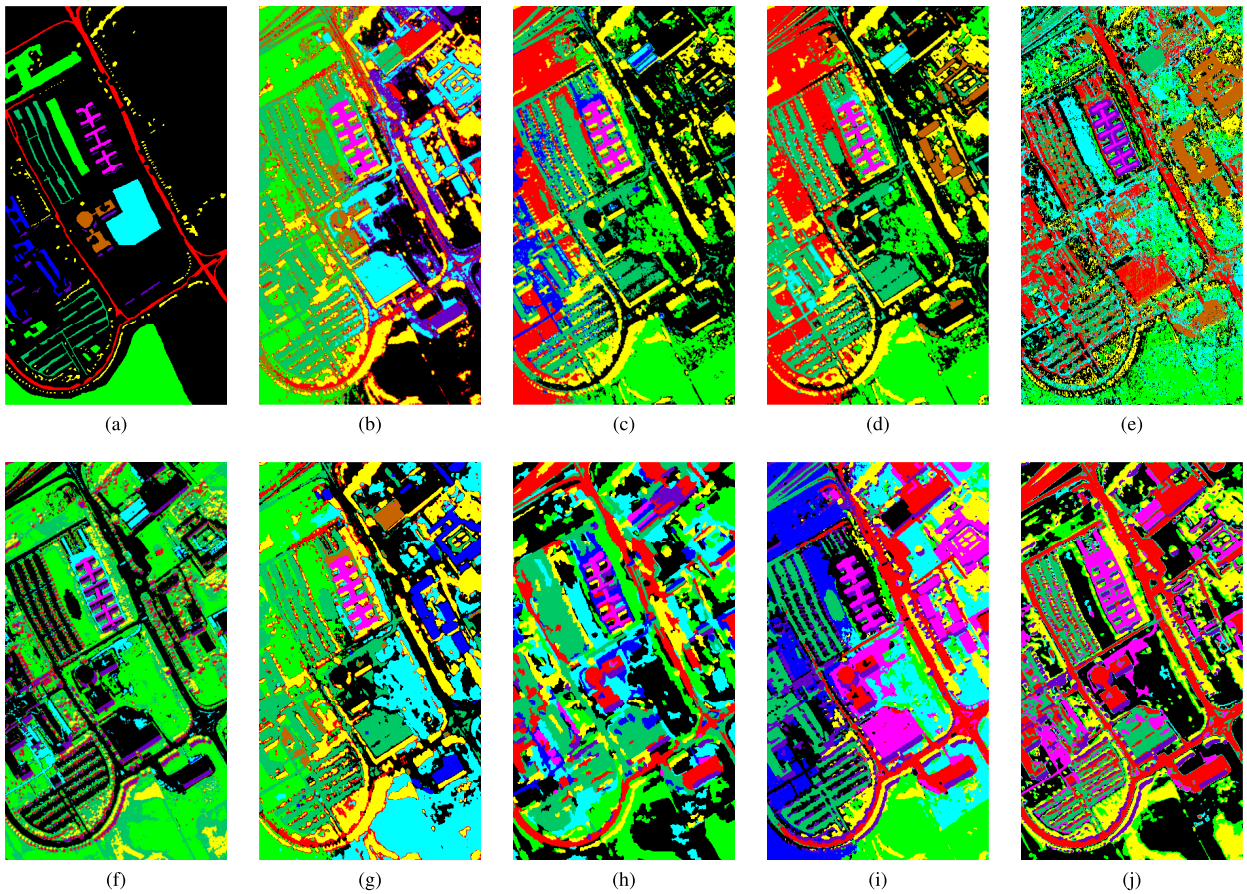


Fig. 3. Clustering maps of the PU dataset. (a) Ground truth. (b) DEC. (c) DEKM. (d) IDEC. (e) KDAE. (f) DCN. (g) DEC-DA. (h) SDST. (i) VTCC. (j) STSDC.

shared, enabling the network to learn diverse hierarchical features at different depths. The number of superpixels is configured as $[300, 1000, 1000, 1100]$ for the ML, PU, YZ, and YC datasets, respectively. The coefficients

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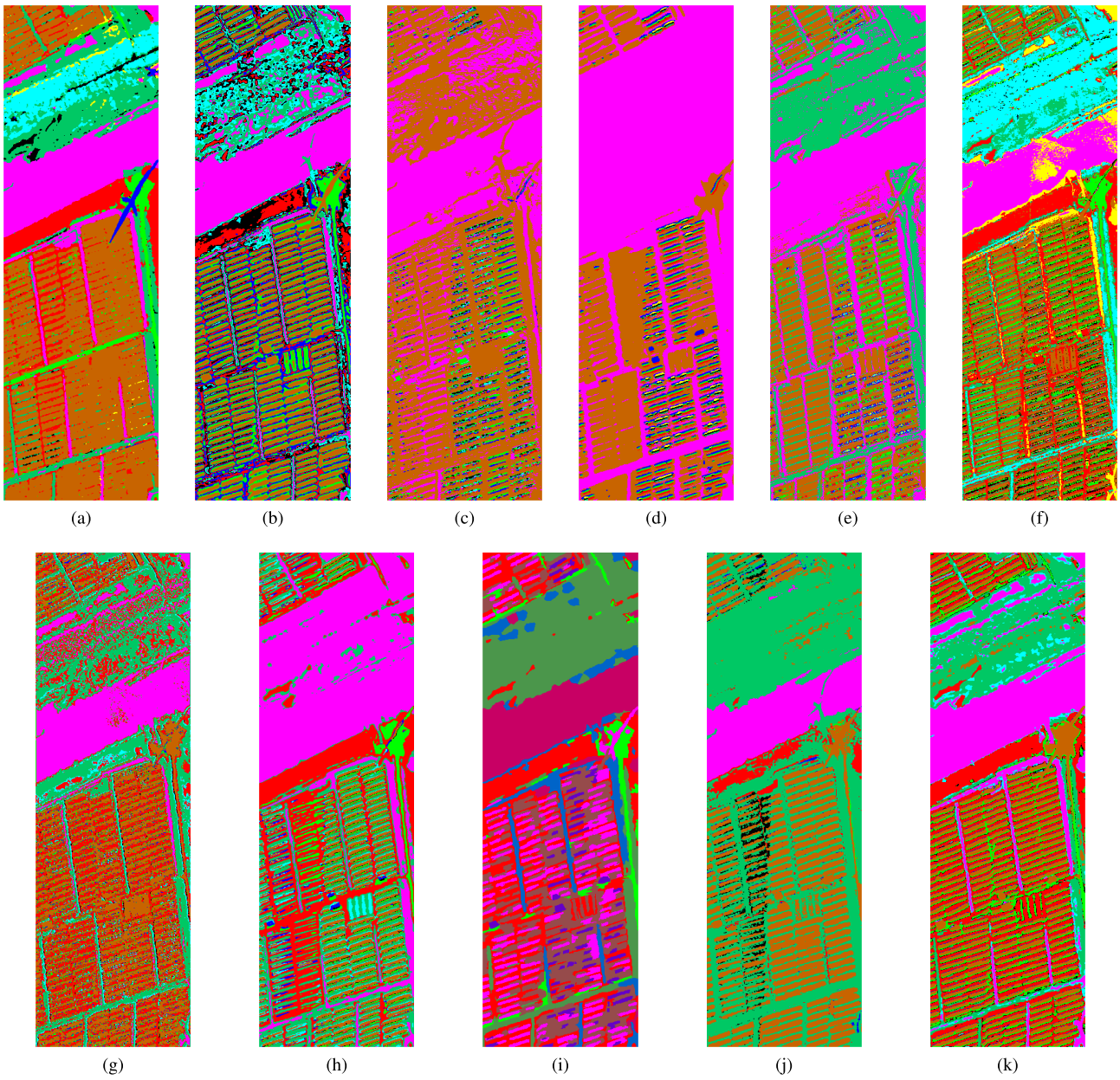


Fig. 4. Clustering maps of the YZ dataset. (a) Ground truth. (b) DEC. (c) DEKM. (d) IDEC. (e) KDAE. (f) DCN. (g) BootSC. (h) DEC-DA. (i) SDST. (j) VTCC. (k) STSDC.

of the loss terms L_{KL}^{MSCB}, L_{KL}^{ST}, and L_{KL}^{cat} are set to $\gamma_1 = 1$, $\gamma_2 = 0.5$, and $\gamma_3 = 0.3$. The model is implemented in PyTorch 2.5.1 with Python 3.10 on Ubuntu 20.04. All experiments are conducted on an NVIDIA A800 (80 GB) GPU. For fair comparison, all baseline methods adopt their officially released implementations and recommended hyperparameters, with minor parameter adaptations to accommodate specific experimental settings.

C. Clustering Performance Comparison

Tables I–IV summarize the quantitative clustering results of the proposed STSDC and all baseline methods on the ML, PU, YZ, and YC datasets, respectively. Figs. 2–5 further present the clustering maps produced by the proposed STSDC and all baseline methods across the four datasets. The best and second-best results are highlighted in bold and underlined. To facilitate a comprehensive analysis, the baseline methods are grouped into four categories: AE-based clustering (DEC, IDEC, DCN, DEKM, and DEC-DA), cluster-specific clustering (KDAE and DWSC), spectral clustering (BootSC), and recent transformer-based clustering (SDST and VTCC).

Table I and Fig. 2 report the results on the ML dataset, which is characterized by highly mixed regions and complex spatial structures. Among the AE-based clustering methods, DEKM achieves the highest ACC. By jointly optimizing deep embeddings and the K-means clustering objective in an end-to-end framework, DEKM preserves intraclass compactness, yielding 77.96% PA on structurally well-defined

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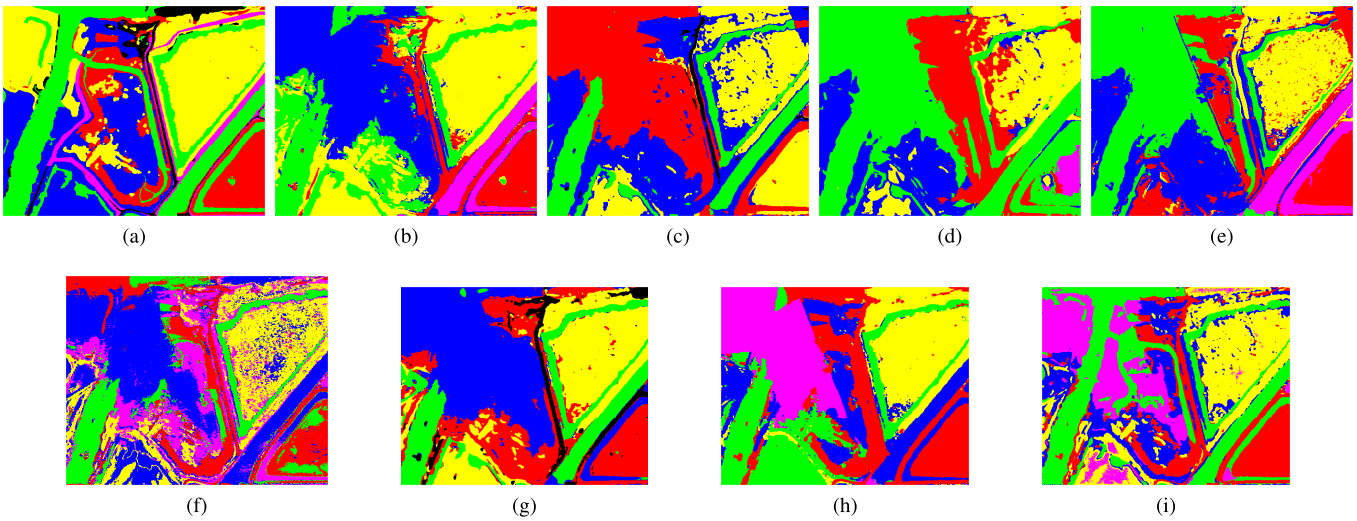


Fig. 5. Clustering maps of the YC dataset. (a) Ground truth. (b) DEC. (c) DEKM. (d) IDEC. (e) KDAE. (f) DCN. (g) DEC-DA. (h) VTCC. (i) STSDC.

TABLE I
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE ML DATASET (%)

Metric	Class	DEC	DEKM	IDEC	KDAE	DCN	BootSC	DEC-DA	DWSC	SDST	VTCC	STSDC
PA	Trees	30.52	53.72	**70.02**	35.89	22.60	33.52	54.50	40.23	28.76	36.47	57.27
	Mostly Grass	0.54	0	1.64	1.36	50.19	89.18	**94.15**	39.65	43.68	76.16	32.67
	Mixed Ground Surface	**90.64**	68.44	73.99	54.08	30.96	16.23	38.52	53.89	59.88	47.22	78.38
	Dirt and Sand	**78.26**	76.40	47.59	0	55.81	5.81	0	15.77	49.95	37.02	0.49
	Road	38.04	77.96	0	46.55	41.78	20.50	56.63	40.86	77.39	80.28	**87.20**
	Water	0	0	0	0	**4.51**	0	0	0	2.36	0	0.64
	Building Shadow	88.36	0.09	0	69.19	85.71	11.42	0.18	23.56	84.68	65.52	**93.95**
	Building	36.84	42.88	33.51	29.66	30.02	**52.42**	43.61	25.53	22.36	50.27	26.47
	Sidewalk	6.06	0.58	1.81	1.81	15.52	3.54	1.59	5.27	25.78	**57.04**	15.23
	Yellow Curb	0	0	0	0	3.28	0	0	0	0	0.55	**6.56**
	Cloth Panels	38.66	46.47	22.30	25.28	0.37	0	47.96	0	**84.76**	0	0
UA	Trees	88.83	92.33	53.46	88.70	80.87	73.97	97.42	94.18	94.97	**99.66**	99.45
	Mostly Grass	22.55	0	1.04	22.66	29.28	23.06	38.34	27.96	27.76	**60.60**	27.10
	Mixed Ground Surface	54.30	42.85	49.38	45.19	58.29	16.67	48.57	22.91	54.82	**73.81**	52.60
	Dirt and Sand	21.31	33.17	**79.65**	0	31.22	6.26	0	22.38	23.77	15.28	0.62
	Road	48.34	27.87	0	16.88	70.72	31.85	23.25	25.95	65.43	**93.00**	63.03
	Water	0	0	0	0	**0.51**	0	0	0	0.18	0	0.35
	Building Shadow	28.75	0.22	0	13.33	42.69	6.89	0.45	13.39	24.59	**61.32**	26.77
	Building	21.40	70.87	69.06	67.90	48.81	39.07	46.56	65.39	63.15	**93.25**	58.05
	Sidewalk	1.94	0.67	2.33	1.60	3.46	2.63	1.50	3.95	10.38	22.30	**30.06**
	Yellow Curb	0	0	0	0	0.13	0	0	0	0	0.02	**4.86**
	Cloth Panels	97.20	79.62	**100**	55.74	0.02	0	96.27	0	36.42	0	0
ACC		40.58	49.58	45.60	34.88	32.36	33.10	48.43	37.19	42.17	49.20	**55.73**
ARI		23.56	32.04	21.82	14.12	14.99	7.30	33.13	12.95	24.18	34.01	**38.45**
AMI		37.66	45.19	36.35	29.07	35.29	16.61	44.44	25.48	40.88	**54.35**	47.78
NMI		37.70	45.22	36.39	29.11	35.32	16.64	44.46	25.52	40.91	**54.37**	47.81
FMI		39.02	48.40	46.06	34.11	29.65	27.08	48.38	31.56	38.63	47.70	**51.48**
FLOPs		928.6G	356.00G	357.50G	508.20G	187.90G	155.80G	750.80G	87.90G	246.00G	1.03T	410.89G
Params		3.25M	133.35K	2.50M	3.56M	1.31M	477.2K	1.07M	0.62M	562.92K	2.30M	2.76M

road, outperforming DEC and IDEC. However, since DEKM relies on spectral similarities without considering the spatial relationships, it fails to discriminate subtle spectral differences in heterogeneous regions, with its PA dropping to 0.09% for building shadow. BootSC utilizes a swapped-prediction mechanism and Sinkhorn balancing to jointly optimize deep spectral clustering from scratch, achieving 33.10% ACC. However, relying on RGB-tailored data augmentations transfers poorly to the ML dataset, causing semantic confusion in heterogeneous regions and yielding 11.42% PA for building shadow. DWSC relies on a weighted k-subspace network to group latent features into low-dimensional subspaces. However, this simple subspace projection mechanism fails to disentangle the nonlinear spectral mixing prevalent in ML, leading to extremely low PAs of 0% and 23.56% for water and building shadow, respectively. Regarding the transformer-based methods, VTCC utilizes a dual-branch transformer with contrastive learning to enhance instance-level discriminability, achieving an ACC of 49.20%. However, VTCC often neglects fine-grained local spatial consistency, leading to misclassifications in local similar regions. SDST adopts a double-structure transformer combining an Autoformer and a Siamese network, achieving a lower ACC of 42.17%. In contrast, the proposed STSDC couples an MSCB with a superpixel-guided semantic transformer to preserve fine-grained structures and associates semantically homogeneous regions, yielding the highest ACC

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TABLE II
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE PU DATASET (%)

Metric	Class	DEC	DEKM	IDEC	KDAE	DCN	BootSC	DEC-DA	DWSC	SDST	VTCC	STSDC
PA	Asphalt	35.26	72.25	72.49	49.01	**98.39**	-	64.82	-	72.88	73.53	97.09
	Meadows	36.60	52.12	49.43	61.55	58.73	-	33.85	-	62.61	49.31	**69.46**
	Gravel	0	**41.88**	0.38	0	0.14	-	0	-	0.81	0.05	0
	Trees	73.89	76.96	81.85	48.92	58.68	-	81.69	-	71.05	73.69	**86.95**
	Painted metal sheets	**100**	100	98.81	75.54	74.72	-	100	-	86.69	100	93.75
	Bare Soil	51.18	0	0.93	46.85	9.72	-	54.25	-	27.00	**57.51**	5.13
	Bitumen	**73.31**	0	0	18.20	0	-	0.08	-	0	0	0
	Self-Blocking Bricks	88.02	80.17	91.69	18.28	0	-	91.20	-	84.33	**97.80**	84.52
	Shadows	0	0	0	27.14	99.58	-	0.63	-	47.10	100	**100**
UA	Asphalt	67.47	53.63	53.04	40.47	40.90	-	56.55	-	70.65	**77.92**	73.30
	Meadows	90.15	80.78	80.52	85.57	74.59	-	90.80	-	79.10	**93.13**	76.83
	Gravel	0	**63.88**	0.10	0	5.36	-	0	-	1.67	0.10	0
	Trees	47.28	55.71	54.02	32.29	53.98	-	49.43	-	32.93	**66.49**	50.00
	Painted metal sheets	**100**	100	100	100	100	-	99.93	-	99.32	48.91	53.59
	Bare Soil	24.29	0	39.50	22.52	84.17	-	23.59	-	33.79	**94.20**	49.81
	Bitumen	**31.46**	0	0	18.17	0	-	1.72	-	0	0	0
	Self-Blocking Bricks	44.41	33.28	40.95	43.45	0	-	40.83	-	46.84	48.32	**65.61**
	Shadows	0	0	0	14.21	**100**	-	0.31	-	85.77	62.67	98.75
ACC		45.73	51.54	49.78	48.56	50.76	-	48.05	-	57.93	58.72	**64.60**
ARI		**48.22**	30.80	32.14	31.31	31.32	-	30.70	-	39.75	45.70	47.36
AMI		29.60	48.07	48.90	38.18	51.68	-	48.84	-	54.03	**64.85**	62.17
NMI		28.25	48.09	48.92	38.20	51.71	-	48.86	-	54.05	**64.86**	62.19
FMI		44.06	46.20	47.36	46.99	49.30	-	45.78	-	53.39	57.50	**59.99**
FLOPs		2.69T	1.03T	1.04T	1.48T	545.50G	-	2.18T	-	467.00G	2.38T	1.40T
Params		3.25M	126.95K	2.50M	3.56M	1.31M	-	1.07M	-	568.03K	2.30M	3.12M

TABLE III
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE YZ DATASET (%)

Metric	Class	DEC	DEKM	IDEC	KDAE	DCN	BootSC	DEC-DA	DWSC	SDST	VTCC	STSDC
PA	luotu	42.27	0.08	0.26	0.05	55.77	35.97	68.83	-	69.65	18.89	**76.85**
	shuini	42.01	4.48	1.08	8.06	49.85	10.49	38.60	-	**50.60**	3.97	39.03
	shuizao	10.23	8.64	6.36	1.59	7.26	8.02	6.01	-	**49.27**	0.76	3.25
	jianzhu	**20.99**	0	0	0	3.60	0.21	0	-	0.71	0.35	0
	caodi	83.62	7.43	**99.66**	94.85	57.28	74.50	86.91	-	71.34	65.82	94.61
	tantu	39.78	0	0	0	**92.98**	4.18	0	-	4.30	0	5.88
	liqing	41.53	**64.41**	59.31	59.29	43.77	38.90	22.40	-	41.95	59.72	41.75
	zacao	26.56	0.01	0.05	84.63	32.97	59.85	22.58	-	77.15	**95.33**	70.21
UA	luotu	41.37	0.93	1.06	0.25	33.96	13.94	34.39	-	39.42	**69.21**	37.99
	shuini	11.31	3.73	1.51	6.08	18.86	5.25	22.60	-	54.59	**80.61**	13.46
	shuizao	0.60	2.59	1.28	0.69	1.35	2.86	2.52	-	**3.20**	1.12	0.96
	jianzhu	**1.45**	0	0	0	0.46	0.18	0	-	0.17	0.10	0
	caodi	87.09	40.88	29.93	52.62	97.51	85.77	42.48	-	96.27	**99.47**	86.42
	tantu	29.38	0	0	0	**34.49**	5.08	0	-	3.92	0	9.41
	liqing	9.37	45.71	76.79	82.50	88.96	84.84	**96.99**	-	87.59	86.59	85.03
	zacao	62.53	0.23	1.52	50.72	45.57	54.56	21.74	-	64.40	31.84	**69.87**
ACC		45.64	39.55	41.66	56.22	48.46	44.56	37.84	-	54.21	55.45	**57.05**
ARI		30.97	7.50	12.58	28.25	29.27	23.71	17.00	-	34.10	19.38	**36.68**
AMI		39.84	15.69	28.27	34.93	44.38	30.54	36.11	-	**47.85**	35.06	47.70
NMI		39.85	15.70	28.28	34.93	44.39	30.55	36.12	-	**47.85**	35.06	47.71
FMI		45.22	38.54	41.75	47.65	44.06	41.02	36.97	-	48.33	44.93	**50.72**
FLOPs		2.60T	990.00G	1.00T	1.42T	560.0G	436.0G	2.10T	-	399.00G	8.23T	1.38T
Params		3.25M	123.75K	2.50M	3.56M	1.40M	476.6K	1.07M	-	587.16K	2.30M	3.10M

of 55.73%. Based on the above analysis, the proposed STSDC method is superior to the other deep clustering methods for the ML dataset both visually and quantitatively.

As observed from Table II and Fig. 3, among the AE-based methods on the PU dataset, DEKM yields the best performance on ACC. Specifically, DEKM maintains intraclass compactness in large homogeneous regions such as Meadows, achieving UA of 80.78%. However, the PU dataset is characterized by highly similar spectral signatures and significant intraclass spatial variations (e.g., painted metal sheets and bitumen). Because DEKM is constrained to a unified global latent space, it cannot disentangle spectral similarity from spatial semantic variation. As a result, DEKM obtains 0% PA on bitumen and 0% PA on shadows, reflecting its limited discriminative capacity in complex regions. KDAE achieves the highest ACC among cluster-specific methods by employing independent encoders. However, KDAE is highly sensitive to the sample imbalance in the PU scene. Consequently, it fails to learn effective representations for the minority gravel class, leading to a 0% PA, which reveals significant instability and overfitting risks under limited data. Comparatively, the advanced transformer baselines VTCC and SDST achieve competitive ACCs of 58.72% and 57.93%, respectively. SDST employs a Siamese transformer to model local relationships,

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TABLE IV
CLUSTERING RESULTS FOR DIFFERENT METHODS ON THE YC DATASET (%)

Metric	Class	DEC	DEKM	IDEC	KDAE	DCN	BootSC	DEC-DA	DWSC	SDST	VTCC	STSDC
PA	Bare soil	50.83	34.94	28.80	44.30	53.87	-	57.74	-	-	**79.17**	73.29
	Water	46.69	53.63	**86.08**	60.90	40.65	-	53.53	-	-	41.60	84.48
	Reed	40.71	47.00	41.59	**47.64**	44.88	-	41.21	-	-	42.05	37.35
	S.alerniflora	**61.21**	42.75	42.42	45.84	44.58	-	55.43	-	-	54.66	51.81
	Grass	3.26	4.92	0	8.63	**30.43**	-	16.82	-	-	14.58	14.05
UA	Bare soil	68.71	16.62	25.77	37.08	58.06	-	42.63	-	-	58.00	**70.02**
	Water	56.62	**93.30**	47.96	47.59	71.87	-	86.84	-	-	51.46	80.25
	Reed	29.44	38.24	**55.18**	45.36	30.37	-	26.40	-	-	51.86	50.02
	S.alerniflora	62.82	54.91	74.34	85.01	65.89	-	69.72	-	-	**85.21**	70.32
	Grass	2.76	**40.97**	0	5.69	9.56	-	25.35	-	-	3.97	4.45
ACC		48.16	43.22	49.06	47.88	44.34	-	50.08	-	-	50.23	**58.41**
ARI		18.23	13.85	15.00	16.66	14.27	-	19.10	-	-	23.01	**35.36**
AMI		23.36	20.14	17.16	21.29	20.00	-	24.27	-	-	26.46	**34.96**
NMI		23.37	20.14	17.16	21.29	20.00	-	24.27	-	-	26.47	**34.96**
FMI		38.08	35.51	38.04	36.32	34.04	-	38.95	-	-	39.95	**49.88**
FLOPs		2.60T	985.60G	1.00T	1.42T	560.00G	-	2.10T	-	-	2.23T	1.45T
Params		3.25M	114.15K	2.50M	3.56M	1.40M	-	1.07M	-	-	2.3M	3.10M

yet remains inferior to STSDC in handling boundary pixels. VTCC provides robust global modeling capabilities but fails to recognize small-scale targets, yielding a severely degraded PA of 0.05% for gravel. In contrast, STSDC integrates MSCB and superpixel-guided attention to jointly model local details and spatial semantic information. Benefiting from this design, STSDC reaches a PA of 100% and a UA of 98.75% on shadows. Therefore, the STSDC proposed in this article is significantly superior to the other typical methods in the PU dataset both visually and quantitatively and achieves the best clustering accuracy with an ACC of 64.60%.

As observed from Table III and Fig. 4, on the YZ dataset, among the cluster-specific methods, KDAE achieves the best clustering performance on ACC. KDAE effectively models the highly mixed and spatially irregular zacao class, achieving a PA of 84.63%. However, because independent encoders of KDAE are trained using only the pixels from their corresponding clusters and lack cross-cluster discriminative constraints, many nonzacao pixels are misclassfication. Among the AE-based methods, IDEC preserves the structure of data distribution through reconstruction loss, achieving a 99.66% PA on the large homogeneous Caodi region. Conversely, the KL-divergence-based soft assignment makes IDEC highly sensitive to local minima on minority or boundary-ambiguous classes such as Jianzhu and Tantu, where the PA drastically degrades to 0%. BootSC employs Sinkhorn balancing to alleviate class-size imbalance, achieving a UA of 84.84% on the Liqing. Transformer-based baselines VTCC and SDST achieve 55.45% and 54.21% ACC, respectively. While VTCC utilizes strong augmentation for contrastive learning, such an augmentation strategy can distort the semantic content of spectral signatures in the complex agricultural scenes of the YZ dataset, resulting in a suboptimal PA of only 0.76% for Shuizao. Compared with SDST and VTCC, the proposed STSDC explores the fine-grained spatial structures and semantic consistency between pixels, achieving remarkable improvements in handling spectrally similar regions. Specifically, by preserving local spatial details, STSDC successfully distinguishes spectrally similar targets, yielding high PAs of 76.85% and 94.61% for luotu and caodi, respectively. Therefore, the proposed STSDC method is superior to the other methods for the YZ dataset both visually and quantitatively and achieves the best clustering results with the best ACC of 57.05%.

As shown in Table IV and Fig. 5, the YC dataset is characterized by wetland vegetation with high spectral similarity. Despite this challenge, the proposed STSDC achieves the highest clustering accuracy with an ACC of 58.41%, significantly outperforming the second-best transformer baseline VTCC by 8.18%. Among the AE-based methods, IDEC effectively identifies large homogeneous regions, achieving the highest PA of 86.08% on the water class. Similarly, while VTCC adopts contrastive learning for enhanced discriminability, its inability to disentangle the subtle spectral variations between bare soil and other vegetation types (e.g., reed, S. alerniflora, and grass), reducing to a 58.00% UA for bail soil. Compared with VTCC, the proposed STSDC obtains the best clustering performance based on the visual effects and accuracies. This further illustrates that the proposed STSDC method can exploit the intrinsic structure characteristics and boost the clustering performance simultaneously for the YC dataset.

To further demonstrate the superiority of STSDC, Tables I–IV show that, except for ACC, STSDC achieves the best performance in ARI, AMI, NMI, and FMI. Meanwhile, although STSDC does not obtain the lowest FLOPs or Params, it remains at a moderate level compared with all baseline methods, achieving an optimal balance between accuracy and computational cost.

D. Ablation Study

1) Effectiveness of MSCB and Transformer: We present ablation studies to validate the effectiveness of the proposed MSCB and semantic transformer on the ML, PU, YZ, and YC datasets. Specifically, four variant models were designed to evaluate the contribution of each component: 1) STSDC-M, a baseline variant containing only MSCB, is used to evaluate the contribution of local feature extraction; 2) STSDC-C, which integrates MSCB and SCA, is used to assess the effect of local semantic continuity modeling; 3) STSDC-G, which integrates

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TABLE V

ABLATION STUDY ON THE EFFECTIVENESS OF MSCB AND SEMANTIC TRANSFORMER (%)

Method	ML	PU	YZ	YC
STSDC-M	42.19	52.11	50.19	46.57
STSDC-C	44.34	52.48	47.45	51.54
STSDC-G	48.19	52.37	46.95	49.91
STSDC-S	49.12	58.33	54.49	54.03
STSDC	**55.73**	**64.60**	**57.05**	**58.41**

TABLE VI

COMPARISON OF DIFFERENT SELF-ATTENTION MECHANISMS (%)

Method	ML	PU	YZ	YC
ScConv [51]	43.48	60.51	54.91	48.17
RefConv [52]	43.63	59.72	44.54	49.25
STViT [53]	46.20	58.50	54.66	56.92
EfficientViT [54]	45.64	45.46	42.45	54.79
STSDC	**55.73**	**64.60**	**57.05**	**58.41**

MSCB and SGA, is used to analyze the effect of global semantic aggregation; and 4) STSDC-S retains the MSCB and the semantic transformer but removes the SE block. The STSDC incorporates all these modules within a unified self-supervised optimization framework to jointly capture multiscale local information and global semantic dependencies. The evaluation metric is overall clustering accuracy (%), and the best performance for each dataset is highlighted in bold. The experimental settings are kept consistent with those described in Section IV-B.

From Table V, it is evident that the proposed MSCB strategy and semantic transformer mechanism effectively address the challenges of fragmented semantics and boundary mixing in HSIs. The MSCB enhances the extraction of fine-grained spectral–spatial details by leveraging multiscale receptive fields and adaptive channel recalibration. Specifically, a comparison between STSDC and STSDC-S indicates that the SE block significantly enhances the discriminability of spectral features. The accuracy drop observed in STSDC-S demonstrates that adaptive channel recalibration is vital for mitigating spectral redundancy, offering complementary benefits beyond what multiscale convolutions alone can achieve. Meanwhile, the semantic transformer, through SCA and SGA, captures both local semantic continuity and global semantic dependencies. The semantic transformer enables the model to focus on semantically meaningful spatial–spectral regions, improving interclass separability and reducing intraclass variability. Consequently, the STSDC demonstrates superior clustering performance across all datasets, validating its ability to integrate multiscale local information with global semantic consistency in a unified self-supervised framework.

2) Comparison of Different Self-Attention Mechanisms: Table VI presents the results of the ablation study comparing different self-attention mechanisms on the ML, PU, YZ, and YC datasets. We evaluated several representative methods including ScConv [51], RefConv [52], STViT [53], and EfficientViT [54], to extract spectral–spatial features. The evaluation metric is overall clustering accuracy (%), and the best performance for each dataset is highlighted in bold.

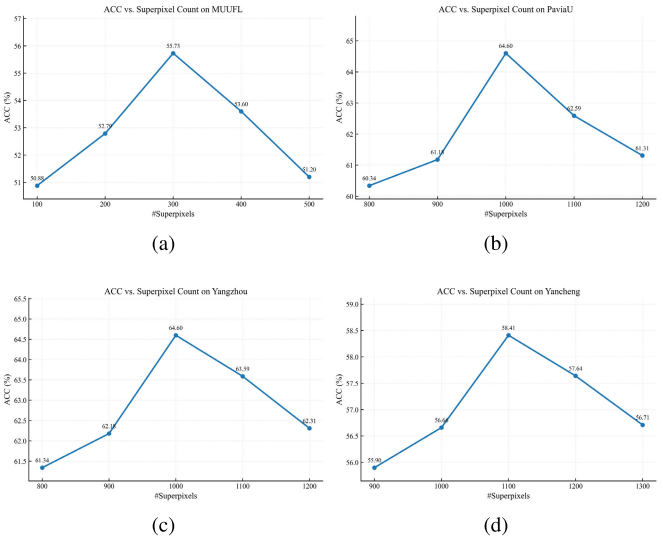


Fig. 6. (a) Effect of superpixel count on MUUFL_ACC. (b) Effect of superpixel count on PaviaU_ACC. (c) Effect of superpixel count on Yangzhou_ACC. (d) Effect of superpixel count on Yancheng_ACC.

Among the baseline methods, ScConv performs competitively on PU and YZ, showing its ability to capture local spectral–spatial features. RefConv achieves stable but lower accuracy across all datasets, indicating limited feature extraction capability. STViT excels on the ML dataset, demonstrating its effectiveness in modeling long-range dependencies, but underperforms on other datasets. EfficientViT performs the worst overall, likely due to insufficient capacity for fine-grained feature extraction and semantic modeling. The proposed STSDC outperforms all baseline methods, achieving 55.73%, 64.60%, 57.05%, and 58.41% on ML, PU, YZ, and YC datasets, respectively. These results validate the effectiveness of STSDC in integrating multiscale local information and global semantic dependencies for HSI clustering.

E. Parameter Analysis

1) Number of Superpixels M: The number of superpixels, denoted as M, is a critical hyperparameter in the proposed STSDC, as it determines the granularity of the semantic primitives for both the SCA and SGA modules. The dataset-specific spatial resolution and complexity lead ERS to favor different typical superpixel sizes to balance region compactness and boundary integrity. Accordingly, we set the number of superpixels $M \in \{100, 200, \ldots, 500\}$ for ML, $M \in \{800, 900, \ldots, 1200\}$ for PU and YZ, and $M \in \{900, 1000, \ldots, 1300\}$ for YC to investigate the clustering performance and verify the optimality of our settings.

The changes in ACC with respect to M are illustrated in Fig. 6. It can be observed that an initial increase in M leads to consistent improvements in clustering performance across all datasets. The initial ACC improvements are attributed to finer superpixel segmentation, which enhances spectral homogeneity within regions, mitigates feature mixing at object boundaries, and provides more precise local priors for the SCA and SGA modules. Subsequently, performance levels off, reaching maxima at $M = 300$ for ML, $M = 1000$ for

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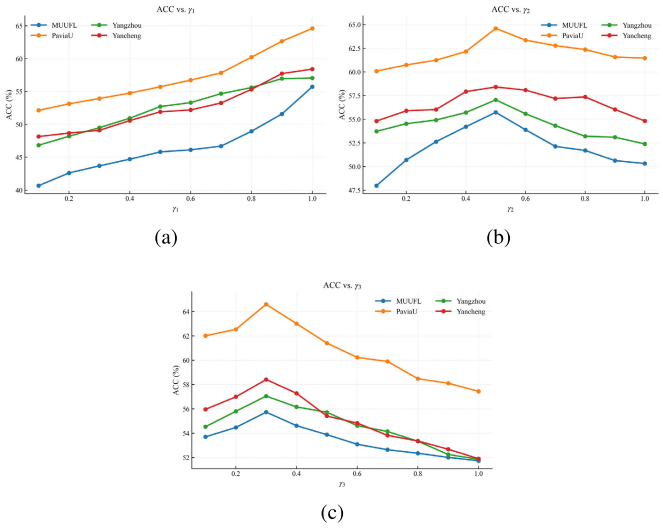


Fig. 7. (a) Effect of coefficient γ_1 on ACC. (b) Effect of coefficient γ_2 on ACC. (c) Effect of coefficient γ_3 on ACC.

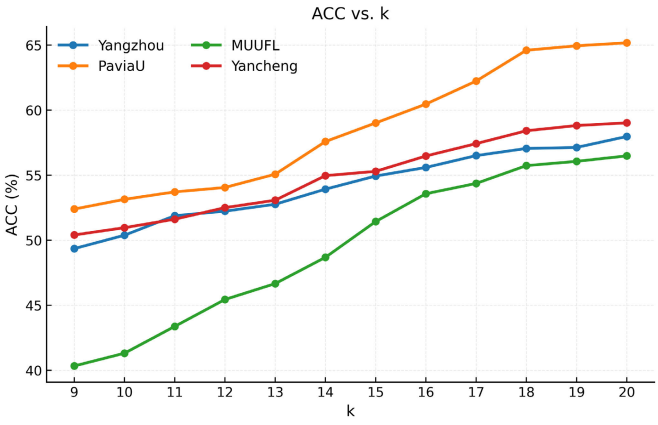


Fig. 8. Effect of number of adjacent superpixels k on ACC.

both PU and YZ, and $M = 1100$ for YC. At these settings, the STSDC achieves an optimal tradeoff between fine-grained spatial structures and semantic integrity.

Beyond the optima, further increasing M reduces ACC across all hyperspectral datasets. The decline arises because excessive segmentation leads semantically homogeneous regions into overly small patches, which hinders the semantic transformer from effectively capturing spatial continuity and modeling global dependencies. Based on the above empirical results, we set the number of superpixels to 300, 1000, 1000, and 1100 for the ML, PU, YZ, and YC datasets, respectively, in all experiments.

2) Parameters γ_1, γ_2, and γ_3: To evaluate the rationale of the hyperparameter configuration in (12), we investigate the sensitivity of the tradeoff weights γ_1, γ_2, and γ_3 using the one-factor-at-a-time sensitivity analysis, with other parameters fixed. The corresponding changes in clustering accuracy with γ_1, γ_2, and γ_3 are presented in Fig. 7.

As illustrated in Fig. 7(a), clustering performance ACC increases steadily with γ_1, attaining a peak at 1.0 across all hyperspectral datasets. The steady rise of ACC with increasing γ_1 indicates that the MSCB branch, which captures fine-grained local structures, is fundamental to feature representation and requires a higher weight. In contrast, Fig. 7(b) and (c) demonstrates that both γ_2 and γ_3 exhibit a convex trend, with performance initially improving as γ_2 and γ_3 increase, but significantly declining once the weights surpass 0.5 for γ_2 and 0.3 for γ_3, respectively. The observed trend reflects that excessive emphasis on the global semantic branch or the fusion constraint can introduce noise or weaken local discriminative details. Accordingly, we adopt the configuration $\{\gamma_1 = 1.0, \gamma_2 = 0.5, \gamma_3 = 0.3\}$ to achieve an effective balance between fine-grained spatial structures and global semantic consistency.

3) Number of Adjacent Superpixels k: The hyperparameter k in (2) defines the effective neighborhood of the SCA mechanism by specifying the total number of selected keys and values, comprising fixed intrasuperpixel tokens and variable adjacent intersuperpixel centers. To identify an appropriate setting, we adjusted k from 9 to 20. Note that $k = 9$ serves as the baseline configuration that includes only intrasuperpixel tokens, whereas each increment beyond nine corresponds to adding one additional adjacent superpixel.

Fig. 8 presents the curve of clustering accuracy ACC against k, showing a clear positive relationship across all hyperspectral datasets. The consistent improvement can be attributed to the incorporation of superpixel-level priors and cross-region context, which enhance semantic continuity and reduce feature mixing at boundaries. As k reaches 18, the rate of improvement begins to level off, suggesting that the most relevant contextual information is effectively represented within this neighborhood size. Further increasing k yields diminishing performance while resulting in additional computational overhead. Therefore, we set $k = 18$ as the default value to achieve an optimal balance between clustering accuracy and computational efficiency.

V. Conclusion

We proposed a novel STSDC to address key limitations of unsupervised HSI clustering, including fragmented semantics and boundary mixing. By integrating an MSCB and a superpixel-guided semantic transformer within a unified self-supervised optimization module, STSDC effectively balances local structure preservation and global semantic consistency. Specifically, the MSCB branch captures fine-grained spectral–spatial details with adaptive channel recalibration, while the semantic transformer introduces SCA and SGA to enhance spatial continuity and model long-range dependencies across disjoint but semantically homogeneous regions. Extensive experiments on benchmark HSIs demonstrate that STSDC achieves state-of-the-art clustering performance, validating its effectiveness and scalability for large HSIs. Future work will investigate an adaptive fusion strategy that dynamically reweights local and global semantic representations across spatial locations to better accommodate the spectral–spatial characteristics present in complex HSIs.

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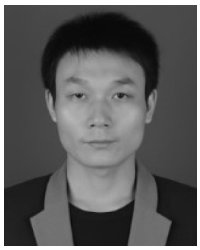
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